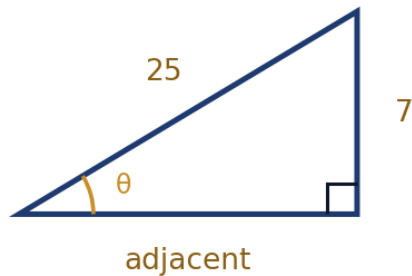


Trigonometry



Right-triangle trigonometry

- 1 In a right triangle, the angle θ has an opposite side of 7 and hypotenuse 25. Find $\cos \theta$ and $\tan \theta$.



- 2 In a right triangle, angle θ has an adjacent side of 8 and opposite side 15. Find $\sin \theta$ and the hypotenuse.
-

Exact values and the unit circle

- 3 State the exact value of $\sin\left(\frac{\pi}{3}\right)$ and $\cos(45^\circ)$.
- 4 State the exact value of $\tan(60^\circ)$ and $\sin(30^\circ)$.
-

Graphs of trigonometric functions

- 5 State the amplitude and period of $y = 4\sin(2x)$.
- 6 State the amplitude and period of $y = 3\cos\left(\frac{x}{2}\right)$.
-

Trigonometric identities

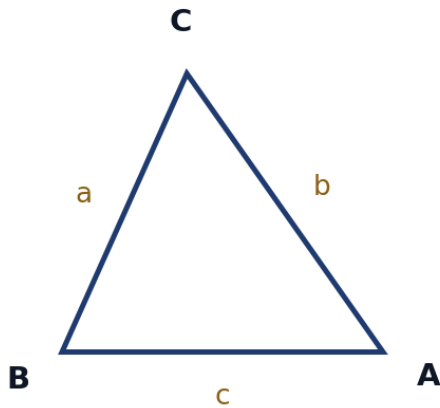
- 7 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using the identity $\sin^2 \theta + \cos^2 \theta = 1$.
- 8 Simplify $\frac{\sin \theta}{1 - \cos \theta} \times (1 - \cos \theta)$.
- 9 Given $\cos \theta = \frac{5}{13}$ with θ acute, find $\sin \theta$ using the identity $\sin^2 \theta + \cos^2 \theta = 1$.
-

Solving trigonometric equations

- 10 Solve $2\sin \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.
- 11 Solve $\cos(2\theta) = -1$ for $0^\circ \leq \theta < 360^\circ$.
- 12 Solve $\tan \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.
-

Law of Sines and Law of Cosines

- 13 In triangle ABC , $a = 9$, $b = 12$, and angle $C = 55^\circ$. Find side c , to the nearest tenth, using the Law of Cosines.



- 14 In triangle ABC , angle $A = 40^\circ$, angle $B = 70^\circ$, and $a = 8$. Find side b , to the nearest tenth, using the Law of Sines.
- 15 In triangle ABC , $a = 7$, $b = 10$, $c = 13$. Find the largest angle, to the nearest degree, using the Law of Cosines.
-

Trigonometric identities for compound angles

- 16 Use the identity $\sin(A + B) = \sin A \cos B + \cos A \sin B$ to find the exact value of $\sin(75^\circ)$, using $75^\circ = 45^\circ + 30^\circ$.
- 17 Use the identity $\cos(A - B) = \cos A \cos B + \sin A \sin B$ to find the exact value of $\cos(15^\circ)$, using $15^\circ = 45^\circ - 30^\circ$.
-

A well-designed word problem: the height of a lighthouse

- 18** This question has two parts. A ship observes a lighthouse at an angle of elevation of 18° . After sailing 500 meters directly toward the lighthouse, the angle of elevation increases to 30° . (a) Let h be the lighthouse's height and d the final distance from its base. Write two equations relating h and d using the tangent ratio. (b) Solve the system to find h , to the nearest meter.